Data Mining – Group Project

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Curricular Unit: Data Mining

Customer Segmentation and Behavioural Analysis

Amazing International Airlines Inc.

Group 40

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1 Executive Summary

This project develops a data driven customer segmentation framework for Amazing International Airlines Inc. (AIAI), with the objective of supporting more targeted marketing actions, loyalty optimisation and improved resource allocation. Through an end-to-end clustering approach, the analysis identifies a set of distinct customer segments that reflect systematic differences in customer value, engagement intensity and seasonal travel behaviour.

From a strategic perspective, the segmentation reveals three broad behavioural dimensions within the customer base. First, a core group of highly engaged and economically valuable customers drives a substantial share of overall activity and revenue, requiring retention-focused strategies that prioritise long term relationship reinforcement, convenience and recognition. Second, a group of moderately engaged and more price sensitive customers displays strong seasonal concentration, representing an opportunity for behavioural conversion through targeted incentives and demand smoothing initiatives. Finally, a smaller set of low engagement or inactive customers exhibits limited recent activity, raising questions around reactivation potential versus operational optimisation.

These insights provide AIAI with a structured basis to transition from mass marketing towards a more personalised and value driven commercial strategy. By aligning incentives and loyalty mechanisms with broad behavioural profiles rather than uniform campaigns, the airline can improve marketing efficiency, stabilise demand across seasons and maximise Customer Lifetime Value. Overall, the proposed segmentation framework translates complex behavioural data into actionable strategic guidance, supporting more informed and sustainable decision-making.

2 Introduction

Building on the exploratory findings from Phase 1, the present stage focuses on preparing the data for customer segmentation and ensuring that the resulting clusters are both meaningful and robust. The analysis relies on two complementary data sources (*flights* and *customer*) and therefore requires a consistent customer-level representation and careful handling of data quality issues identified previously.

Data consistency and duplicates have started with basic consistency checks to ensure a reliable customer-level structure. Duplicate records were removed from the *flights* dataset, while a small number of repeated identifiers in the *customer* dataset were discarded due to their negligible impact. The ***Loyalty#*** identifier was subsequently used as the common index across datasets to guarantee consistency throughout the analysis. In addition, inconsistencies identified during Phase 1 were corrected to support meaningful temporal and behavioural analysis: flight count variables were enforced as integers to remove non-meaningful decimal values, and date-related variables were converted to an appropriate datetime format to support subsequent temporal analyzes.

Handling of missing values. Missing values were observed exclusively in the *customer* dataset and were handled using a feature wise strategy guided by business interpretation. Missing income values were assumed to reflect customers who chose not to disclose this information and were therefore imputed using the median to preserve robustness. Missing values in ***Customer Lifetime Value*** were treated analogously, using median imputation to reduce the influence of extreme values and ensure stability in customer value profiling. The ***CancellationDate*** variable contained the largest number of missing values, which were assumed to correspond to active customers. These cases were imputed with **31-12-2021**, representing the end of the observation period.

Univariate outlier treatment. Given the skewed distributions and long right tails observed during Phase 1, extreme values were further addressed through a univariate outlier treatment. Boxplot inspections of numerical variables (Fig. 1 and Fig. 2) indicated that outliers were concentrated in the upper tail of the distributions. A conservative capping strategy was therefore applied to limit the influence of extreme observations while preserving the overall structure and variability of the data.

Preparation for clustering to ensure comparability across variables and avoid scale dominance, numerical features were rescaled to a common range. In parallel, categorical variables were encoded to support cluster profiling and interpretation. High cardinality identifiers, such as customer names and postal codes, were excluded due to their limited analytical value, whereas location related attributes (country, city, province and state) were retained given their relevance for geographical profiling. A dedicated subset containing only numerical features was also defined for clustering algorithms that require numerical inputs.

Scope of subsequent methodology. More complex stages of the workflow, including the revised feature engineering process and the identification and treatment of multivariate outliers, are discussed in detail in the Methodology section, where the underlying assumptions, definitions and parameter choices are fully documented.

3 Methodology

3.1 Feature Engineering

Feature engineering was performed to transform the available information into meaningful and informative variables suitable for customer segmentation. Given the two complementary datasets, the process was structured into three components: features derived exclusively from the *flights* dataset, features derived from the *customer* dataset and features constructed using information shared across both sources. Several transformations were revised and extended relative to Phase 1 in order to better capture behavioural heterogeneity and customer value.

3.1.1 Feature Engineering – Flights Dataset

A set of behavioural features was engineered from the *flights* dataset to capture travel activity, engagement and temporal preferences. These features were grouped into five categories: **summary**, **recency**, **distance-based**, **seasonal** and **ratio** features. Together, these groups provide aggregated measures of overall activity, recent engagement, travel intensity, temporal patterns and relative behavioural behaviour. A detailed description of the engineered flight-related features is provided in Table 1.

3.1.2 Feature Engineering – Customer Dataset

Six features were engineered from the *customer* dataset to capture customer value, engagement status and socioeconomic characteristics. Particular emphasis was placed on transformations of ***Customer Lifetime Value (CLV)***, given its relevance for segmentation and its potentially non-linear distribution.

The feature ***Months_Enrolled*** measures the duration of each customer's relationship with the loyalty program up to December 2021 for active customers. To improve robustness and interpretability, ***Income_Bins*** and ***CLV_Bins*** were created by discretizing income and CLV into three ordinal levels. Finally, the binary feature ***Is_Active*** distinguishes between active and cancelled customers, capturing the current status of the customer–company relationship. A complete description of the engineered customer-related features is reported in Table 2.

3.1.3 Merging the Datasets

The *flights* and *customer* datasets were merged at the customer level to enable a unified analysis combining behavioural and profile information. During this step, a small inconsistency was identified: 20 customer identifiers (*Loyalty#*) appeared only in the *customer* dataset and shared identical enrollment and cancellation dates. These observations were removed, after which the datasets were successfully merged into a single dataset containing both the engineered features and the original customer attributes.

3.1.4 Feature Engineering with Both Datasets

One cross dataset feature, ***Frequency***, was constructed by computing the ratio between the total number of flights taken by each customer and the number of months since enrollment. This feature captures overall customer activity while accounting for differences in customer tenure.

3.2 Multivariate Outliers Treatment

Multivariate outliers were identified using DBSCAN, a density-based clustering algorithm well suited for detecting observations that do not belong to dense regions of the feature space and that may adversely affect clustering results. The *MinPts* parameter was defined following the common rule of thumb $MinPts = 2\times$dimensionality, resulting in $MinPts = 58$. The neighborhood radius parameter (ε) was selected using the k-distance graph, with the elbow point corresponding to $\varepsilon = 0.7$.

Based on these parameters, DBSCAN identified 156 observations as noise, which were treated as multivariate outliers and removed from the dataset. This procedure resulted in the retention of 97.89% of the original observations for subsequent clustering analyses.

3.3 Feature Selection

Two filter based methods were applied for feature selection, resulting in a final set of 12 features. First, Spearman's correlation was analyzed using heatmap visualizations (Fig. 3) to identify redundant variables through monotonic relationships. Features with correlation values greater than 90% were removed. Additionally, features exhibiting very weak correlation with all others were excluded, as they do not share common structure with the dataset and may introduce noise in an unsupervised setting.

A univariate variance analysis was then conducted, keeping only features with variance close to 2% or more to ensure sufficient dispersion for clustering. The final heatmap (Fig. 4) presents the Spearman correlation of the selected features.

Despite a low correlation with other variables, *Income* was retained due to its sufficient variance and relevance for capturing customer's economic capacity and overall value. For seasonal information, only ratio based features were kept, as they provide a normalized representation of customer behaviour, enabling the identification of seasonal preferences independently of customer scale.

3.4 Defining Perspectives

To refine the segmentation strategy, the analysis was structured around two complementary perspectives: ***Customer Value & Engagement*(VE)** and ***Patterns & Preferences*(PP)**. This separation allows customer value and engagement to be analysed independently from travel behaviour, preventing high activity customers from dominating behavioural insights.

The first perspective focuses on economic value, loyalty and behavioural intensity, combining financial indicators with measures of customer activity and engagement to capture both longterm value and recent interaction. In contrast, the second perspective captures customers travel styles and lifestyle related behaviours independently of their overall value, relying on ratio-based and variability measures to emphasise relative preferences, seasonal habits and companionship patterns.

3.5 Perspective 1: Customer Value & Engagement Clustering

3.5.1 Hierarchical Clustering

For this model, a comparative analysis of hierarchical clustering methods showed that Ward's linkage consistently achieved higher R^2 values than the Complete, Average, and Single linkage techniques, indicating superior explanatory power (Fig. 5). Analysis of the Ward dendrogram (Fig. 6) revealed clear separation patterns, suggesting a small number of dominant cluster groupings.

Although the Elbow Method indicated an inflection at three clusters and the Silhouette Score (Fig. 7) peaked at $k = 2$, these solutions produced overly broad and imbalanced segments. A six cluster solution was therefore selected, as it yielded a more balanced distribution of observations and captured finer distinctions in customer behaviour.

3.5.2 K-Means Clustering

The K-Means solutions were evaluated using inertia and the mean Silhouette Score (Fig. 8). The inertia curve does not display a clearly defined elbow, although a reduction in marginal gains is observed after $k = 3$, indicating that low values of k lead to overly coarse partitions, while additional clusters continue to capture meaningful within-cluster variance.

The Silhouette Score peaks at $k = 2$ and gradually decreases as k increases, reflecting the expected trade off between cluster separation and segmentation granularity. While smaller values of k achieve higher silhouette scores, they result in highly aggregated solutions with limited analytical usefulness.

Given the prioritisation of explained variance, a six cluster solution was selected as a compromise between model fit and interpretability. This configuration achieves higher R^2 values, produces more balanced clusters and offers finer differentiation between customer profiles, while remaining consistent with the structure identified through Hierarchical Clustering.

3.5.3 Density-Based Clustering

Mean Shift

Mean Shift was applied with bandwidth tuning across multiple quantile values. Small quantile changes led to large variations in the number of clusters, indicating smooth behavioral transitions rather than well defined density peaks. The selected configuration (quantile = 0.065) produced five clusters with an R^2 of 0.5159. However, the low Silhouette Score (0.2351) and highly imbalanced cluster sizes indicate weak separation and substantial overlap. Overall, despite reasonable global fit, the strong dependence on bandwidth tuning and poor cluster separation suggest that Mean Shift is not suitable for robust segmentation in this feature space.

DBSCAN and HDBSCAN

For DBSCAN, parameter tuning was guided by the k-distance plot ($k = 12$), indicating an inflection region around $\varepsilon \approx 0.14$, while *minPts* was defined using the heuristic rule $2\times$ the number of features. Multiple configurations were tested to balance segmentation quality and cluster size distribution. The selected solution ($\varepsilon = 0.12$, *min_samples* = 35) produced five clusters and identified 1,086 observations as noise, achieving an R^2 of 0.4006 and a Silhouette Score of 0.1654, indicating weak cluster separation.

HDBSCAN was subsequently applied as a hierarchical extension of DBSCAN to better handle density variations, following a similar tuning strategy. The selected configuration (*min_cluster_size* = 100, *min_samples* = 20) consistently yielded low R^2 values (around 0.39) with limited sensitivity to parameter changes. Although five clusters were identified (excluding noise), the resulting cluster structure was highly imbalanced, suggesting that density-based approaches are not well suited for modelling customer value and engagement patterns in this feature space.

Gaussian Mixture Model (GMM)

This modell was applied using a probabilistic clustering approach. The AIC and BIC criteria (Fig. 9) stabilised at five components, yielding an R^2 of 0.5456, the highest among models in this family. However, the low Silhouette Score (0.0915) reflects the soft clustering nature of GMM and the overlap between Gaussian components.

The estimated parameters, including cluster means and covariance matrices ($5 \times 6 \times 6$), enabled a detailed mathematical characterisation of cluster dispersion and relative weight within the population.

3.5.4 Self-Organizing Maps (SOM)

Self-Organizing Maps were trained using a 15×15 grid, selected to balance resolution and error convergence. After training, the topographic error (TE) decreased from 0.9518 to 0.0147, indicating strong topology preservation, while the quantization error (QE) converged to a stable value of 0.1444.

The component planes (Fig. 10) show clear spatial organisation. Higher values of *Total_Flights*, *Distance_KM* and *Points_Redeemed* are co-located in the same region of the map, whereas *Recency* peaks in a distinct area associated with lower engagement. *Months_Enrolled* aligns with engagement related variables, while *Income* displays structured variation, indicating that behavioural engagement and economic value are related but not fully overlapping. These patterns are supported by the U-Matrix and hit-map (Figs. 11, 12).

The SOM weight vectors were clustered using both K-Means and Ward's methods (Fig. 13), yielding identical structures; K-Means was selected due to its higher explained variance. The final SOM–K-Means solution identified six clusters, achieving an R^2 of 0.7548 and a Silhouette Score of 0.2920.

3.6 Perspective 2: Patterns & Preferences Clustering

3.6.1 Hierarchical Clustering

Following the same hierarchical clustering strategy adopted in the first perspective, Ward's linkage method was selected due to its superior R^2 performance (Fig. 14). The analysis of the Ward dendrogram (Fig. 15) and the mean Silhouette Score (Fig. 16) supported a solution with a small number of clusters as the most appropriate partition for this perspective.

However, a five cluster solution was ultimately selected, as it provided a more balanced distribution of observations across clusters while achieving higher overall explanatory power. Although solutions with fewer clusters exhibited slightly higher silhouette values, they resulted in overly aggregated and less informative segments.

3.6.2 K-Means Clustering

Following the same K-Means clustering procedure described in the first perspective, higher Silhouette values were observed (Fig. 17) for low values of k. However, these solutions resulted in broader and less differentiated clusters.

A five cluster solution was therefore selected to increase segmentation granularity and explain a larger share of variance ($\mathrm{R}^2 = 0.60$), while achieving a balanced distribution of observations across clusters. Despite lower silhouette separation, the resulting clusters remained interpretable and better aligned with actionable marketing objectives.

3.6.3 Density-Based Clustering

Mean Shift

Using the same density based approach as before, Mean Shift exhibited strong sensitivity to the bandwidth parameter, with small variations leading to substantial changes in the number of clusters, suggesting smooth behavioural transitions rather than well defined density peaks.

The selected configuration (quantile = 0.09, bandwidth $\approx$ 0.256) produced seven clusters, achieving an R^2 of 0.5826 and a Silhouette Score of 0.3774. Despite this performance, cluster sizes were highly imbalanced, with one dominant cluster and several very small segments.

DBSCAN and HDBSCAN

This methods were evaluated using the same tuning strategy applied in previous analyses. For DBSCAN, the k-distance plot suggested an inflection region around $\varepsilon \approx 0.13$, with neighbouring values tested to improve stability, while *min_samples* was adjusted to balance cluster structure and noise proportion. The selected configuration ($\varepsilon = 0.13$, $min\_samples = 50$) identified five clusters and 1,737 noise points. Although this solution achieved a reasonable trade-off between interpretability and noise handling, the resulting clusters were highly imbalanced, with one dominant cluster.

HDBSCAN was subsequently applied following the same tuning rationale. The selected configuration ($min\_cluster\_size = 50$, $min\_samples = 200$) produced four clusters and 2,684 noise points. Despite this compromise, the solution was characterised by a high proportion of low-confidence assignments and unstable cluster membership. Overall, consistent with the first perspective, density-based approaches proved unsuitable for robust segmentation in this feature space.

Gaussian Mixture Model

Once more, this model was initially fitted with three components (Fig. 18) and later expanded to seven components, as the results improved substantially. This improvement was observed not only in the R^2 score but also in the distribution of observations across clusters. The final solution achieved an R^2 of 0.5854 and a reasonably balanced cluster structure.

3.6.4 Self-Organizing Maps

As in the first perspective, a 15×15 SOM grid was used to capture granular behavioural patterns. Training quality was confirmed by a strong reduction in topographic error from 0.9720 to 0.0318, indicating good topology preservation, while the quantization error converged to 0.1471, reflecting adequate representation of the input space.

The component planes (Fig. 19) reveal clear seasonal preferences and structured behavioural patterns. Higher summer travel is spatially aligned with increased flight variability and a higher proportion of trips with companions, indicating more irregular and leisure oriented behaviour concentrated in specific periods. In contrast, spring travel follows a complementary pattern, being inversely associated with autumn travel and only moderately aligned with variability and companionship, suggesting more regular travel behaviour. These patterns are supported by the U-Matrix and Hit-Map (Figs. 20, 21), which highlight well defined regions with heterogeneous observation densities.

The SOM weight vectors were clustered using K-Means and Hierarchical methods (Fig. 22). While both produced comparable structures, K-Means consistently achieved higher R^2. The final SOM–K-Means model identified five clusters, achieving an R^2 of 0.6180 and a Silhouette Score of 0.2344, yielding stable and interpretable travel behaviour profiles.

4 Results and Validation

4.1 Perspective 1: Model Comparison

The final clustering model for this perspective was selected based on a comparative analysis of R^2, Silhouette Score and cluster structure (Fig. 23). Among the evaluated approaches, K-Means and Self-Organizing Maps achieved the highest R^2 values, with K-Means obtaining a marginally superior result. In addition, K-Means also achieved a higher Silhouette Score, indicating better overall cluster separation.

Furthermore, the K-Means solution produced a balanced distribution of observations across clusters and clearly distinguishable segments, while density-based methods resulted in highly imbalanced structures. Considering both quantitative performance and interpretability, K-Means was selected as the most appropriate model for this perspective.

4.2 Perspective 2: Model Comparison

The model comparison followed the same evaluation criteria adopted in the first perspective (Fig. 24). All clustering methods achieved similar R^2 values, indicating comparable explanatory power, with the main differences arising in terms of the Silhouette Score. Density-based approaches, such as DBSCAN and HDBSCAN, exhibited higher silhouette values; however, their solutions were characterised by highly imbalanced clusters, limiting their practical usefulness for segmentation.

In contrast, partition-based methods produced more balanced and stable cluster structures. In particular, the combination of Self-Organizing Maps (SOM) followed by K-Means resulted in a six cluster solution with well populated and clearly distinguishable segments. Despite not achieving the highest silhouette score, this approach offered a better balance between variance explanation, cluster distribution and interpretability. Consequently, the SOM-based K-Means solution with $k = 6$ was selected as the most appropriate model for this perspective.

4.3 Final Clustering Model and Analysis

The final clustering solution was selected by evaluating multiple model combinations and merged configurations, guided by R^2, cluster size distribution and interpretability. K-Means was selected for the **VE** perspective and Self-Organizing Maps (SOM) for the **PP** perspective.

Hybrid segments were subsequently merged using Hierarchical Clustering with Ward's linkage. Different dendrogram cut levels were analysed (Fig. 25), and a solution with **six merged clusters** was selected as the best compromise between separation, stability and interpretability.

Across PCA, t-SNE and UMAP projections (Figs. 26, 27, 28), customer segments appear partially overlapping and connected by smooth transitions rather than sharply separated. Linear projections emphasise dominant behavioural gradients, while non-linear methods reveal dense cores with fuzzy boundaries, indicating a mix of niche and broader heterogeneous groups. The consistency of these structures across all projections supports the stability of the final segmentation and reflects the continuous nature of customer behaviour.

4.4 Feature Importance

Feature relevance was assessed using two complementary approaches: an R^2 based importance measure and a Decision Tree classifier. Although feature rankings differ between methods due to their distinct definitions of importance, both analyses consistently highlight key drivers related to recent activity, travel intensity, income and variability in flying behavior. This convergence supports earlier exploratory and clustering results and confirms that the final segmentation is driven by meaningful and interpretable behavioral dimensions.

4.5 Outlier Reclassification

Outliers identified using DBSCAN display extreme yet internally coherent behavioural patterns, typically associated with lower activity and stronger seasonal preferences. As shown in Fig. 29, most of these observations lie at the periphery of the PCA projection, clearly separated from the dense population core.

Although a small subset appears embedded in central regions of the two dimensional space, this is a consequence of dimensionality reduction. The projection does not preserve the full density structure of the original high dimensional feature space, where these observations remain isolated according to the DBSCAN criterion. Their behavioural distinctiveness therefore justifies exclusion from the main clustering stage, allowing the final segmentation to focus on stable and representative customer profiles.

4.6 Customer Segment Profiling

Cluster profiling was performed using metric, binary and categorical features to characterise the final customer segments. Metric variables provided the strongest discrimination across clusters and therefore formed the core of the analysis, while binary and categorical attributes were used to validate and contextualise the observed behavioural patterns.

The resulting segmentation is unbalanced. Cluster 1 is the largest segment, representing approximately 46% of the customer base, followed by Cluster 0 with around 28%. The remaining clusters (2, 3, 4 and 5) are substantially smaller, each accounting for less than 10% of the population. Clusters 3 and 4 correspond to niche segments with strong seasonal characteristics, while Cluster 5 captures a distinct group of inactive customers.

Metric feature profiling reveals well-defined behavioural patterns across clusters. Cluster 0 is characterised by higher income, above-average travel activity and balanced seasonality, indicating a financially valuable and consistently engaged segment. Cluster 1 exhibits the highest travel intensity, long programme tenure and high variability in flying behaviour, positioning it as the most loyal and operationally critical segment. Cluster 2 comprises moderately active customers with lower income, limited engagement and a pronounced summer travel bias. Clusters 3 and 4 are driven by strong autumn and winter seasonality, respectively, reflecting specific travel preferences rather than high frequency. Finally, Cluster 5 is distinguished by extreme recency and negligible travel activity, corresponding to an inactive or churned segment.

5 Strategic Recommendations

Based on the behavioural segmentation, a set of targeted strategic recommendations was developed with the objective of transitioning from mass marketing to a more personalised and value driven approach. These strategies aim to maximise Customer Lifetime Value (CLV), improve retention and optimise marketing efficiency by aligning incentives with the specific behavioural drivers of each segment.

For **Cluster 0 - The Elite**, a high value and affluent segment, the strategic focus should move away from monetary incentives towards exclusivity, convenience and status recognition. Dedicated concierge support, priority services and partnerships with premium hospitality providers can reinforce loyalty by reducing friction and enhancing perceived prestige.

Cluster 1 - The Loyal Traveler, representing the largest and most loyal segment, requires strategies centred on long term relationship reinforcement. Loyalty mechanisms that reward tenure, facilitate family travel and acknowledge historical engagement are particularly relevant for safeguarding this group against competitive switching.

Cluster 2 - The Casual Traveler presents an opportunity for behavioural conversion. Given its strong summer orientation and lower engagement, targeted seasonal incentives and gamified challenges can be employed to stimulate off peak travel and encourage more regular programme usage throughout the year.

Clusters 3 - The Autumn Traveler and **Cluster 4 - The Winter Enthusiast** correspond to highly seasonal niche segments. For these customers, value creation should focus on curated, season specific offerings and the removal of logistical barriers, such as tailored cultural packages for autumn travel and enhanced support for winter sports related journeys.

Finally, **Cluster 5 - The Ghost** represents a segment with no recent activity. This group can either be addressed through low cost reactivation campaigns based on short term incentives or strategically excluded from active marketing efforts to improve operational efficiency and database quality.

Overall, this segmentation driven strategy enables the alignment of marketing actions with distinct behavioural profiles, ensuring that resources are allocated to the segments where they generate the highest strategic and financial return.

6 Conclusion

This project implemented an end-to-end customer segmentation pipeline, encompassing data preprocessing and feature engineering, multi model clustering, cluster merging, profiling and visualisation. The combined use of hierarchical, centroid-based, density-based and neural approaches ensured a balance between robustness, interpretability and statistical performance, with visual analyses confirming segment coherence.

From a business perspective, the resulting segmentation reveals clear differences in customer value, engagement and seasonal travel behaviour, providing AIAI with a solid basis for targeted marketing actions, personalised loyalty strategies and more efficient resource allocation.

Future work could extend this approach through temporal modelling or hybrid frameworks combining clustering and predictive models, as well as by incorporating additional behavioural or contextual data to enhance the long-term strategic value of the segmentation.

7 References

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A Appendix A

A.1 Univariate Outliers Treatment Appendix

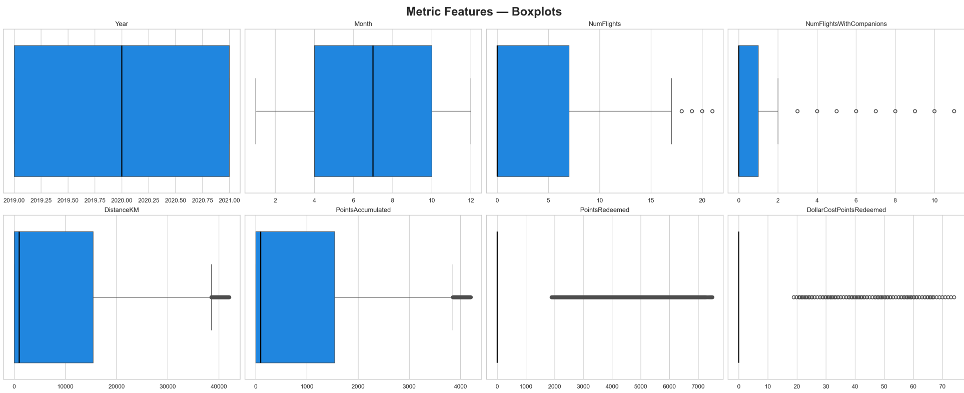

Figure 1: Univariate Outliers Analyses – Flights

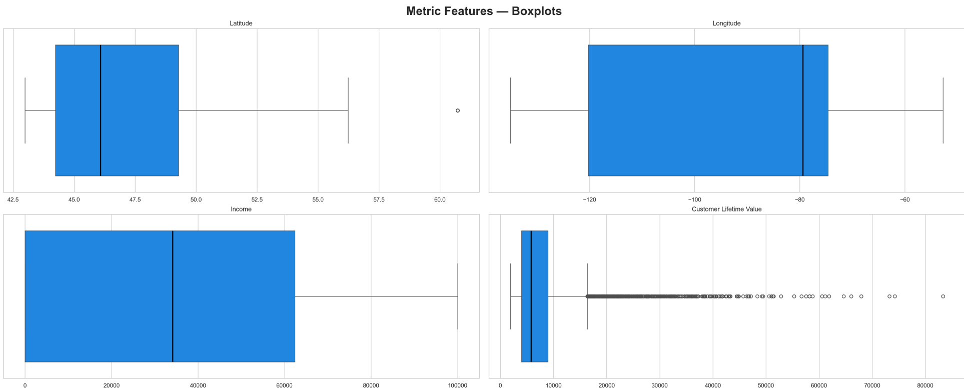

Figure 2: Univariate Outliers Analyses – Customer

A.2 Feature Engineering Appendix

Table 1: Engineered features for the flights dataset

Feature	Description
Total_Flights	Total number of flights taken by each customer
Total_Flights_With_Companions	Total number of flights taken with companions
Total_DistanceKM	Total distance travelled (in kilometers)

Feature	Description
Total_Points_Accumulated	Total loyalty points accumulated
Total_Points_Redeemed	Total loyalty points redeemed
Flights_Variance	Variance in the number of flights taken
Did_Travel	Binary indicator of whether the customer travelled
Solo_Travels	Number of solo trips taken
Only_Travel_With_Companions	Customers who only travelled with companions
Recency	Months since the most recent flight (Dec 2021)
Distance_Per_Flight	Average distance travelled per flight
Winter_Flights	Number of winter flights
Spring_Flights	Number of spring flights
Summer_Flights	Number of summer flights
Autumn_Flights	Number of autumn flights
PreferredSeason	Season with the highest number of flights
Total_Flights_With_Companions_Ratio	Ratio of flights with companions
Winter_Flights_Ratio	Ratio of winter flights
Spring_Flights_Ratio	Ratio of spring flights
Summer_Flights_Ratio	Ratio of summer flights
Autumn_Flights_Ratio	Ratio of autumn flights

Feature	Description
log_CLV	Logarithm of Customer Lifetime Value
relation_inc_clv	Ratio between CLV and income
Months_Enrolled	Number of months enrolled in the loyalty program
Income_Bins	Income grouped into three levels
CLV_Bins	Customer Lifetime Value grouped into three levels
Is_Active	Binary indicator of customer status

Table 2: Engineered features for the customer dataset

A.3 Feature Selection Appendix

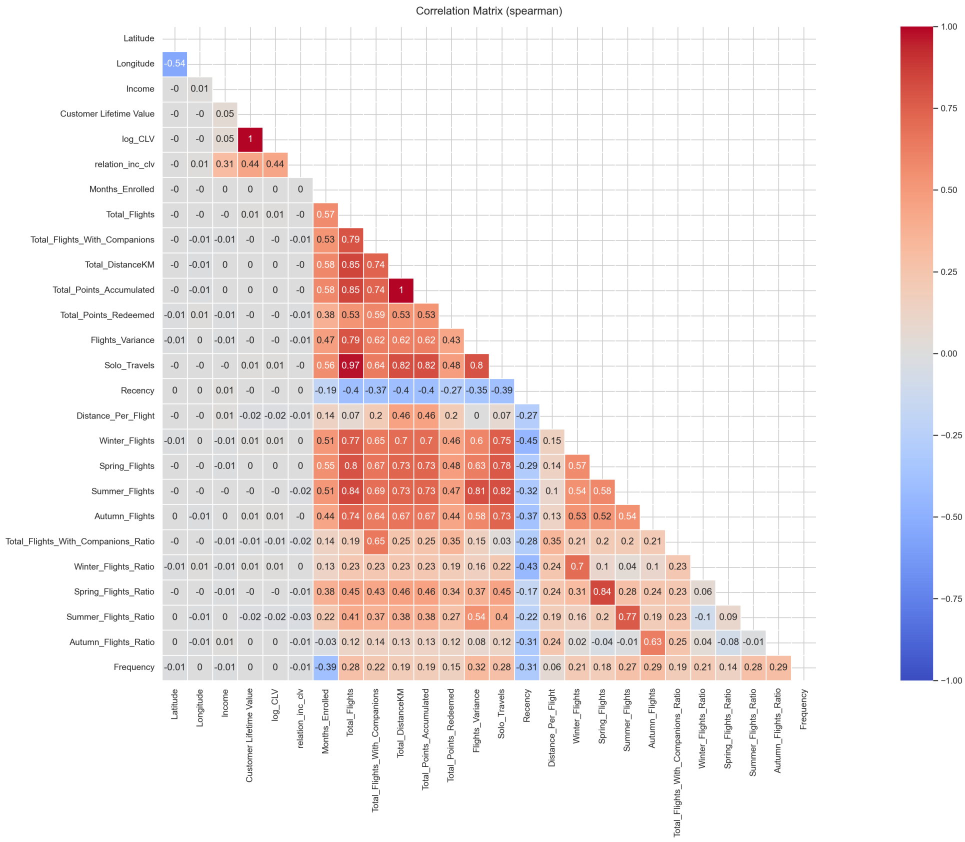

Figure 3: Correlation Matrix (spearman)

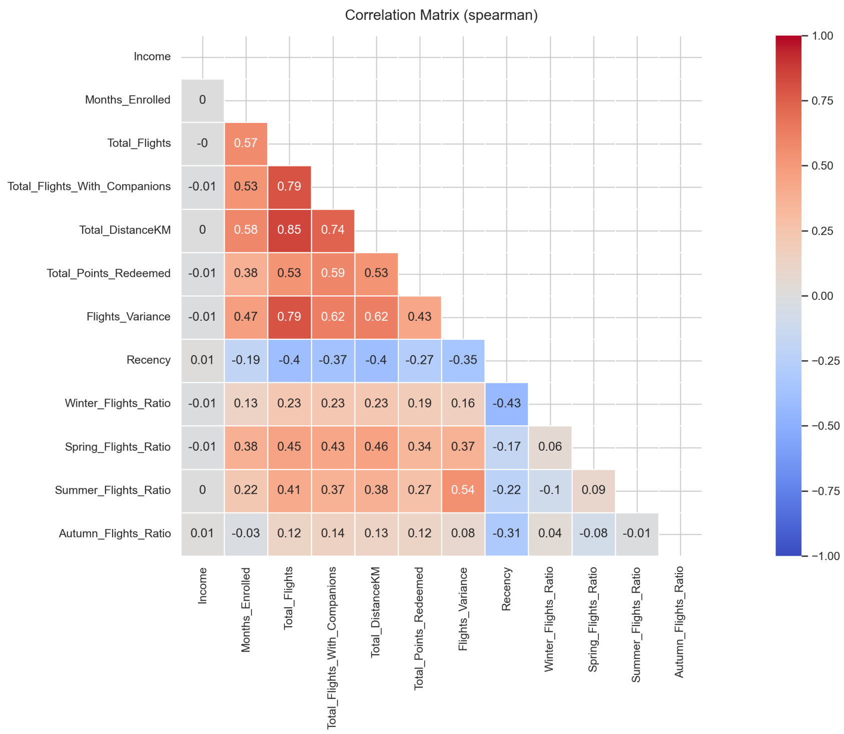

Figure 4: Correlation Matrix of the selected features

A.4 Clustering Appendix

Figure 5: Comparative analysis of R^2 per Hierarchical Method - VE Perspective

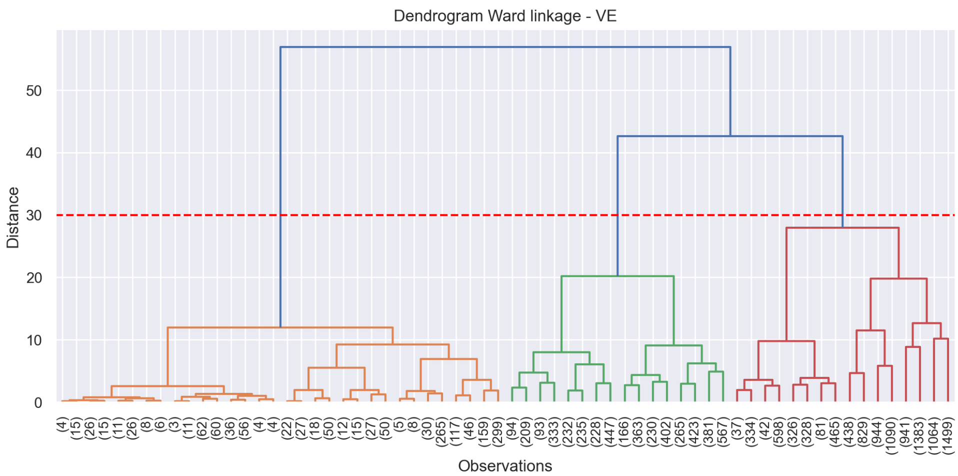

Figure 6: Dendrogram (Ward linkage) - VE perspective

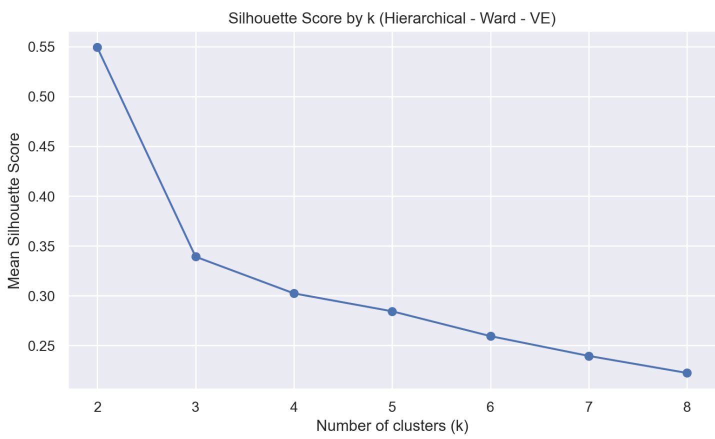

Figure 7: Silhouette Score (Hierarchical Clustering) - VE perspective

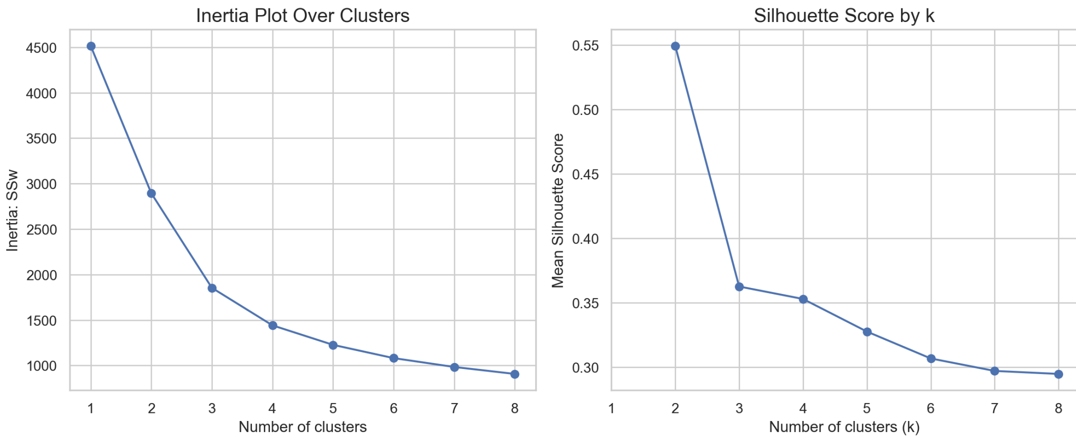

Figure 8: Inertia and Silhouette Score Plot (K-Means) - VE perspective

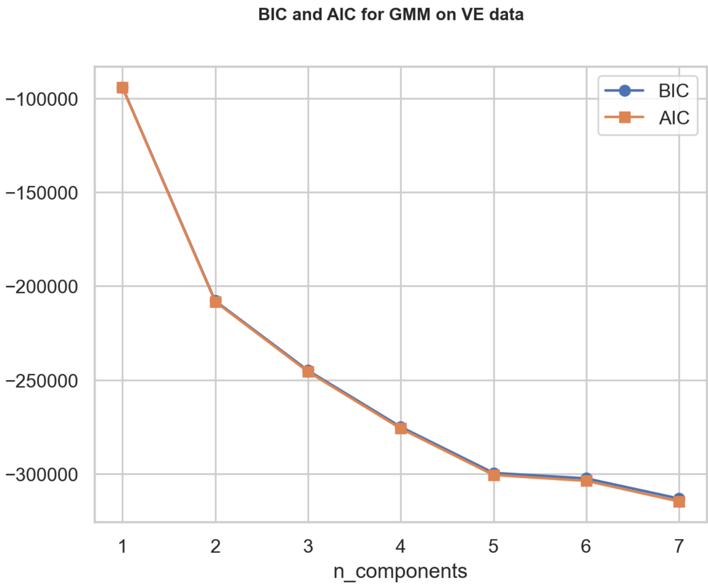

Figure 9: Comparative analysis using BIC and AIC criteria - VE perspective

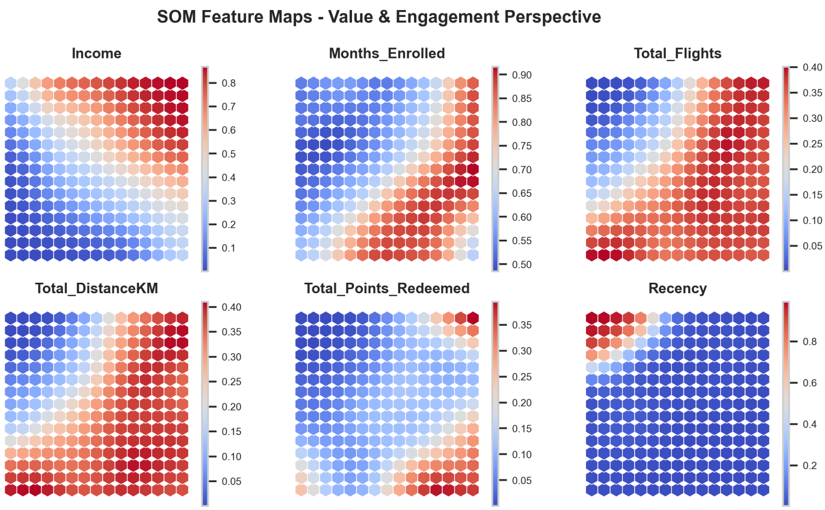

Figure 10: Component Planes (SOM) - VE perspective

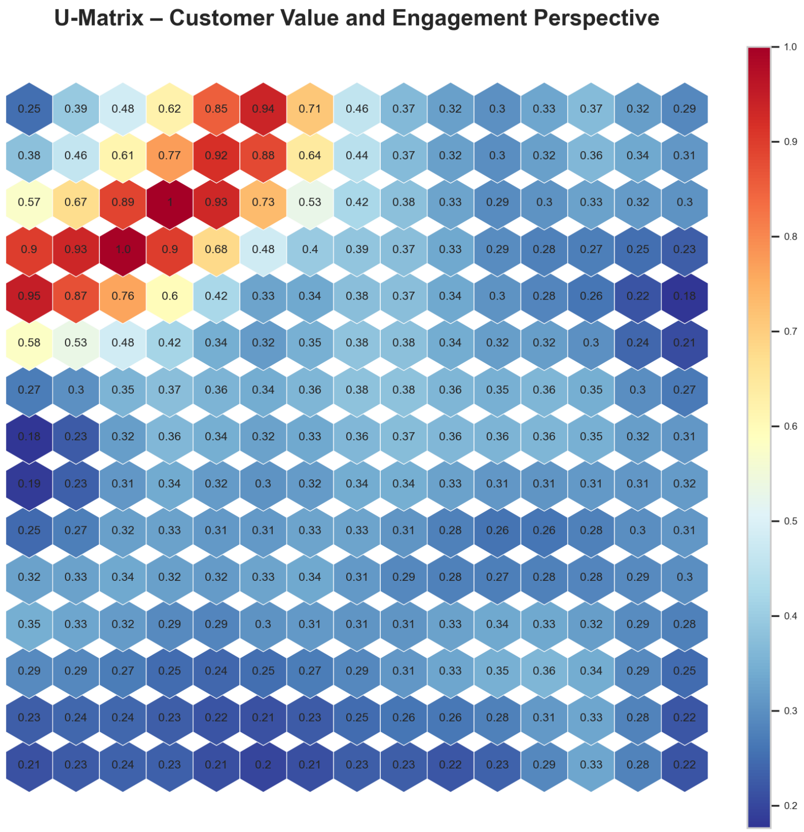

Figure 11: U-Matrix (SOM) for VE perspective

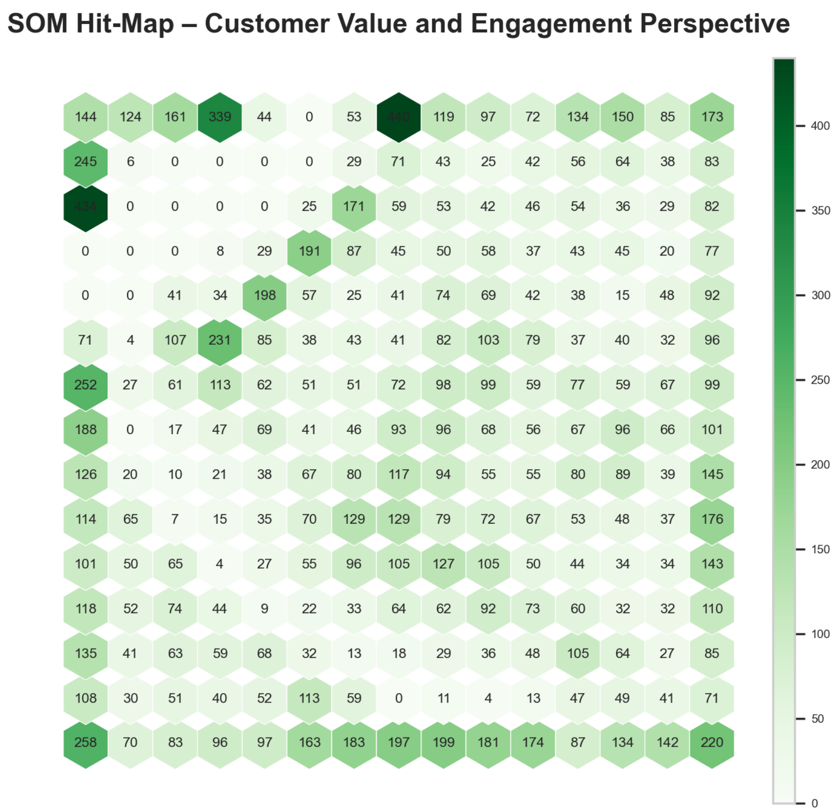

Figure 12: Hit Map (SOM) for 1st perspective

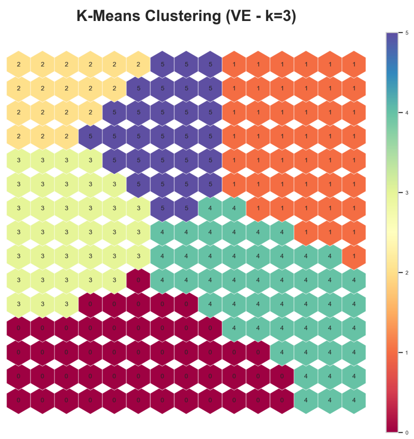

Figure 13: Comparative analysis of SOM with K-Means and HC - VE perspective

Figure 14: Comparative analysis of R^2 per Hierarchical Hethod - PP Perspective

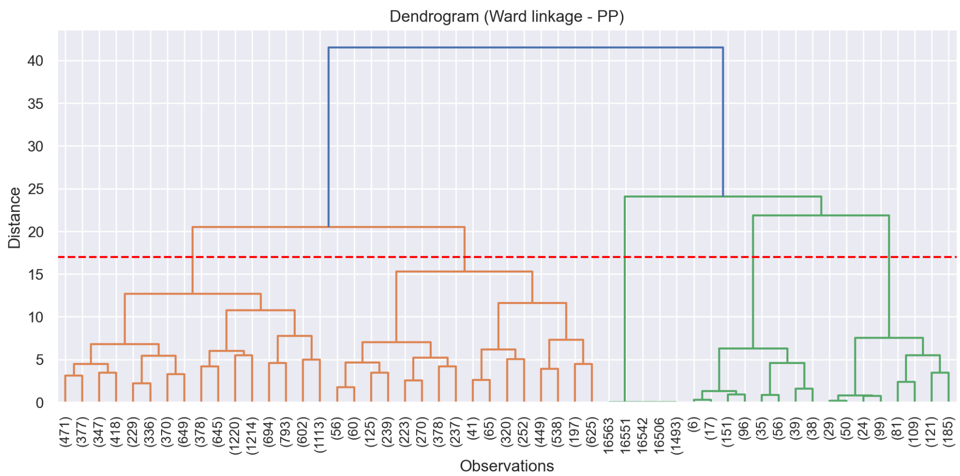

Figure 15: Dendrogram (Ward linkage) for 2nd perspective

Figure 16: Silhouette Score (HC) for 2nd perspective

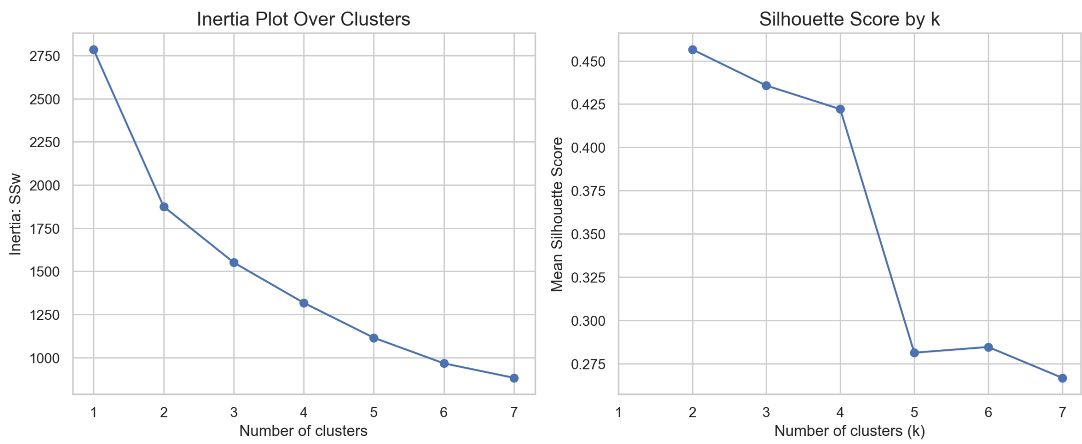

Figure 17: Inertia plot and Silhouette Score (K-Means) for 2nd perspective

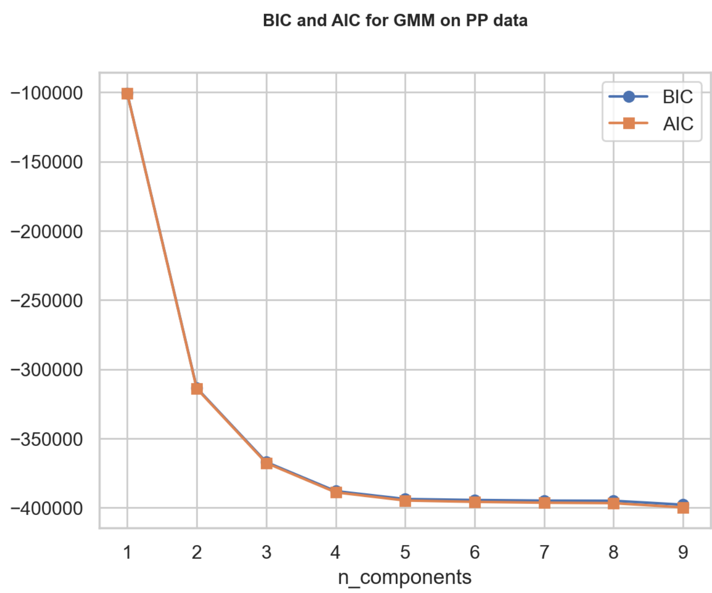

Figure 18: Comparative analysis using BIC and AIC criteria - 2nd perspective

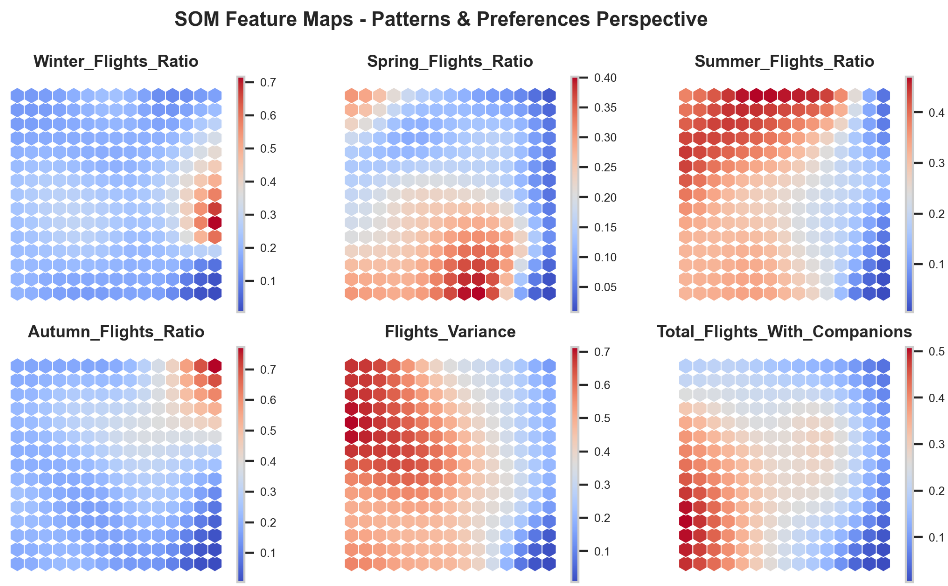

Figure 19: Component Planes (SOM) - 2nd perspective

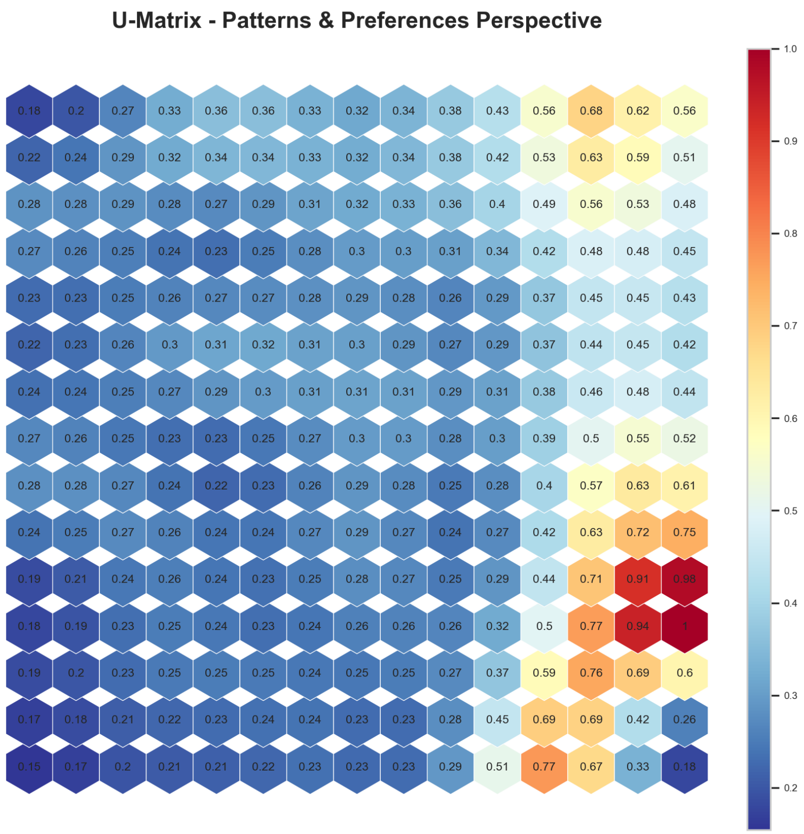

Figure 20: U-Matrix (SOM) for 2nd perspective

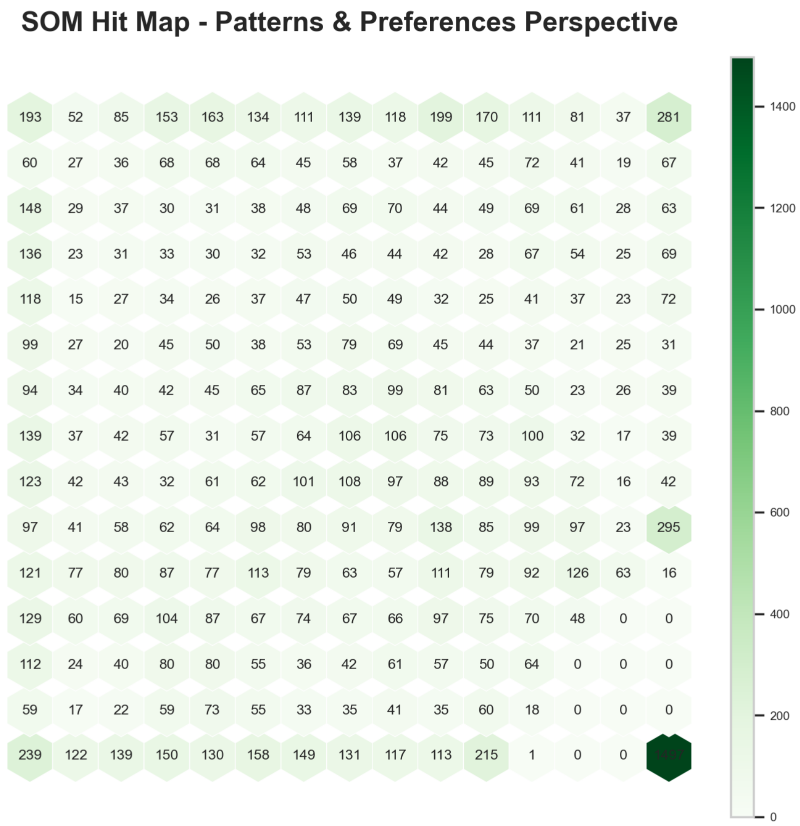

Figure 21: Hit Map (SOM) for 2nd perspective

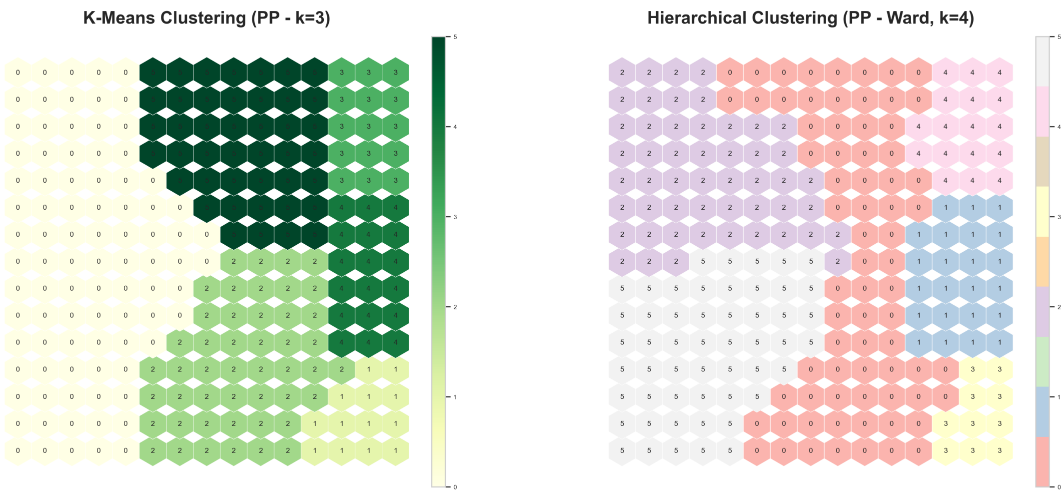

Figure 22: Comparative analysis of SOM with K-Means and HC - 2nd perspective

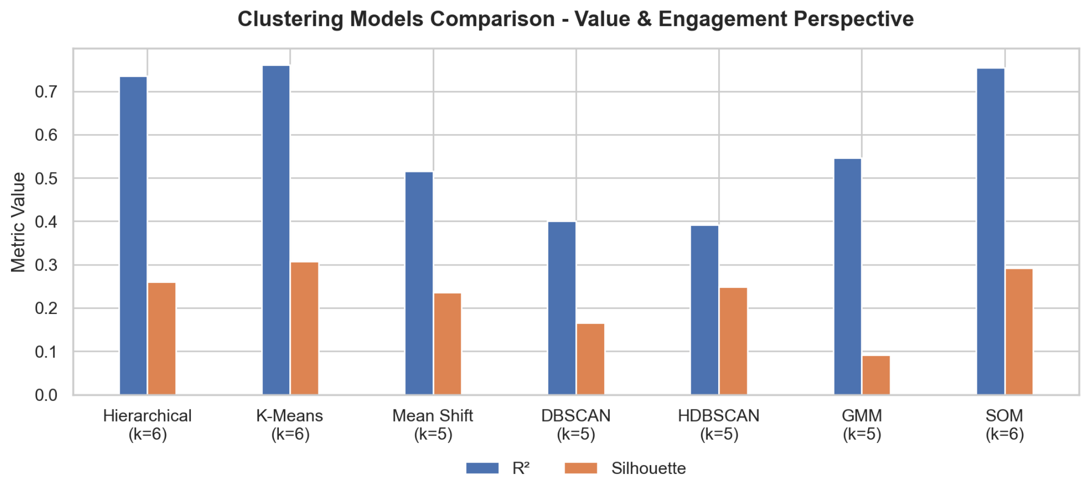

Figure 23: Comparing models for VE perspective

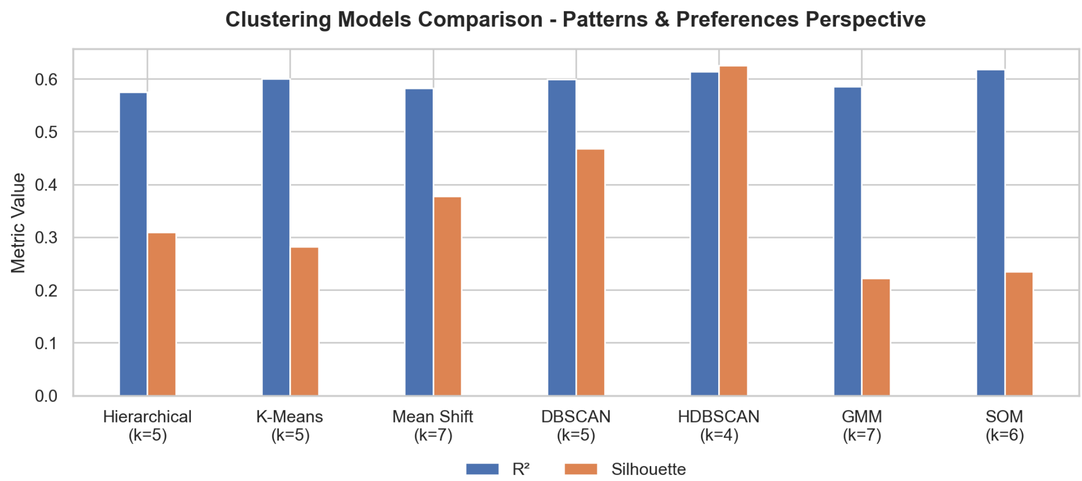

Figure 24: Comparing models for 2nd perspective

A.5 Merged Clusters Appendix

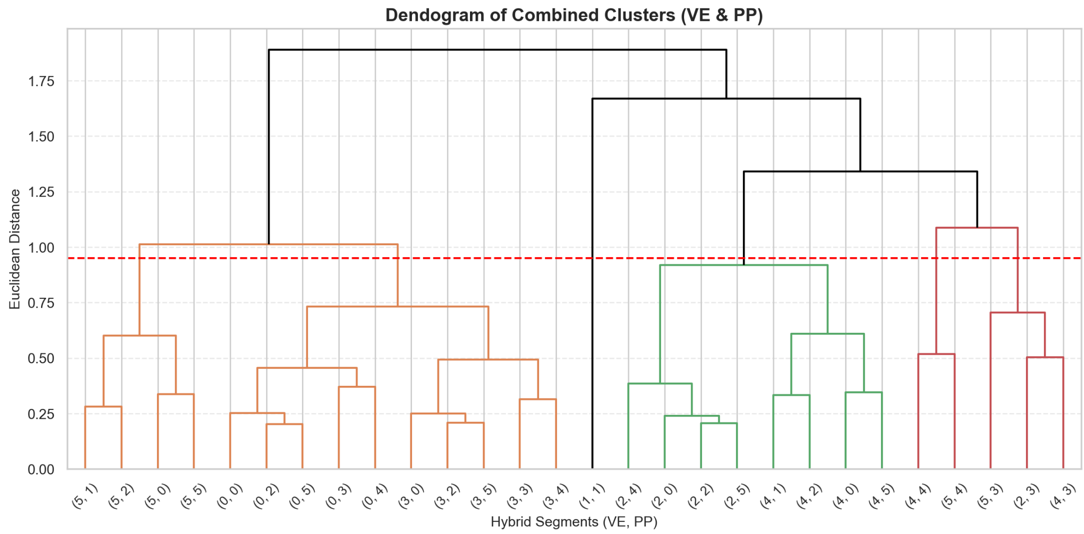

Figure 25: Dendrogram of the merged segmentation

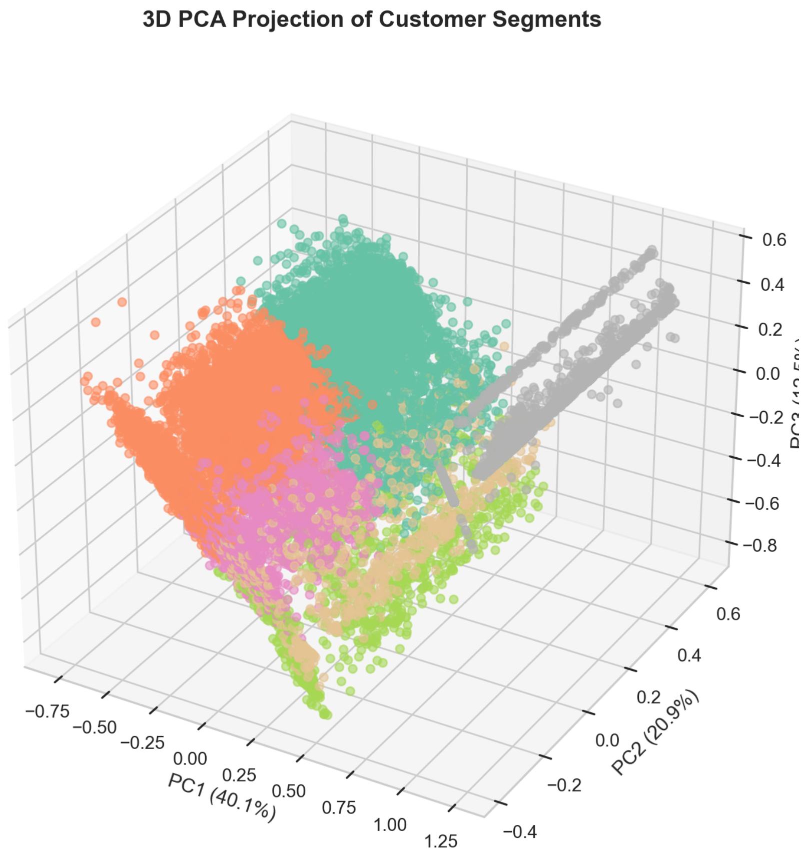

Figure 26: 3D projection of customer segments in the space of principal components

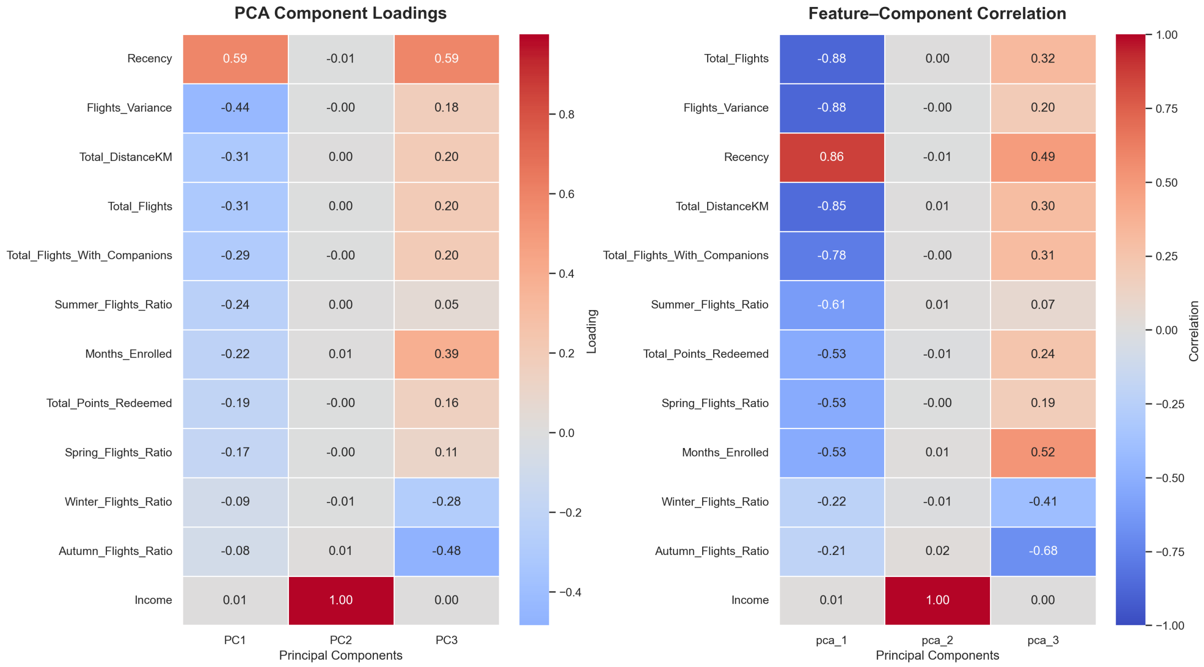

Figure 27: PCA component loadings (left) and feature–component correlations (right)

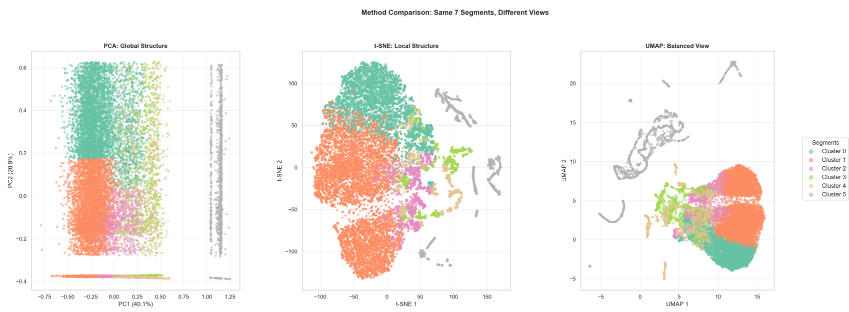

Figure 28: Clustering solution visualised using PCA, t-SNE and UMAP

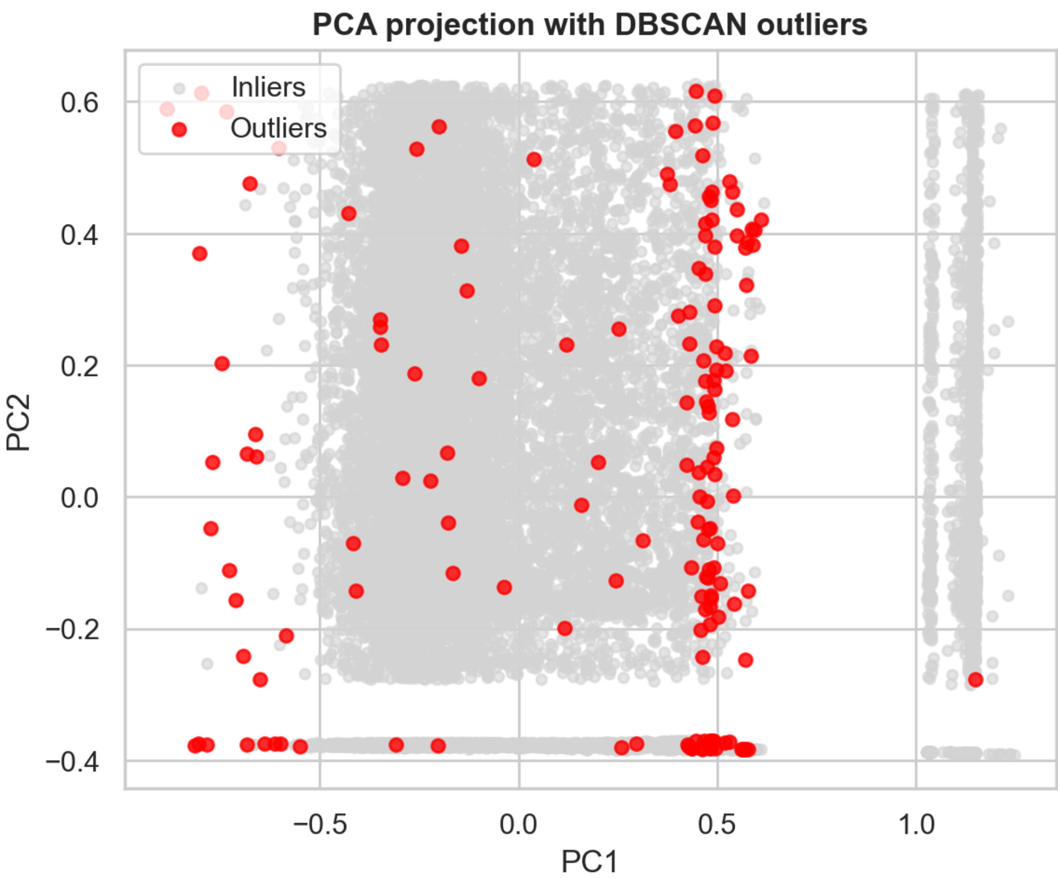

Figure 29: DBSCAN outliers identified in the PCA-transformed feature space

A.6 AI Usage Statement

Several Large Language Models were used throughout the project to support productivity and improve the quality of both the analytical workflow and the final deliverables. ChatGPT was the primary tool employed, complemented by Perplexity for theoretical clarification, Gemini for cross-validation of explanations and GitHub Copilot for additional coding support.

In particular, ChatGPT played a central role in the development, refinement and debugging of complex Python code used for data analysis, custom functions and advanced visualisations. This included the design of non trivial plotting functions, clustering evaluation graphics and profiling visualisations that would have been difficult to implement independently due to their structural and logical complexity. While the underlying methodologies and modelling decisions were fully understood and guided by the author, ChatGPT was instrumental in translating these ideas into correct, efficient and reusable code.

Additionally, these tools were used to assist with code optimisation, error resolution, interpretation of results and decision making, as well as to improve the clarity, coherence and academic tone of the written report. All outputs generated with AI assistance were critically reviewed, adapted and validated to ensure correctness, consistency and alignment with the project objectives.

A.7 Contribution Statement

All members contributed to collaborative discussions, critical decision making and ideation.

Maria Fonseca (Student ID: 20250380)

- Revision and Improvement of EDA and Feature Engineering
- Scaling, Encoding and Multivariate Outliers Detection
- Feature Selection
- Hierarchical Clustering, K-Means and Gaussian Mixture Model Clustering
- Comparison between Models and Selection of best Model
- Profiling
- Feature Importance
- Video

Ana Macedo (Student ID: 20250405)

- Revision and Improvement of EDA and Feature Engineering
- Defining Perspectives
- Self-Organizing Maps Clustering
- Slides for Video
- Video

Lourenço Silva (Student ID: 20250453)

- Revision and Improvement of EDA and Feature Engineering
- Mean Shift, DBSCAN and HDBSCAN Clustering Methods
- Profiling for Categorical Features
- Merged Clusters Visualizations
- Video

A.8 Responsibility Statement

We, the group members listed above, certify that this report represents our original analytical work and interpretations. While AI tools were used as specified above, all insights, conclusions, and recommenda- tions are the result of our independent analysis and critical thinking. We take full responsibility for the accuracy and quality of this submission.